WORKED SOLUTION

STEP I 2004, Question 10

Mechanics • Time-distance

STEP I 2004, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

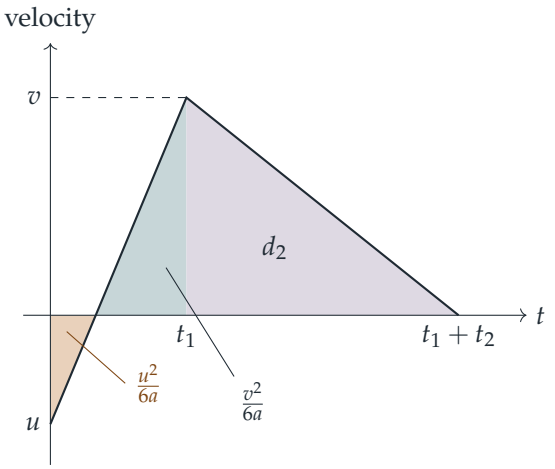
step.maths.org/questions/04-s1-q10

The key idea

Everything here is constant acceleration, so $v^2 = u^2 + 2as$ and $v = u + at$ do all the work. The one trap is the word *distance*. When $u < 0$ the particle starts by moving backwards, stops, and only then speeds up forwards, so the distance d_1 is the sum of two pieces rather than a single displacement. A velocity–time graph makes this vivid: distance is the total area between the graph and the time axis, counting areas below the axis as positive.

Solution

Take the direction of the final velocity v as positive. In the first stage the acceleration is $+3a$ (it takes the velocity from u up to $v > u$); in the second it is $-a$, until the velocity is 0. Throughout, $a > 0$.



The graph shows the case $u < 0$: the small triangle below the axis is the backward

journey.

The second stage (both cases). From speed v to rest with deceleration a, the particle never reverses, so

$$0 = v^2 - 2ad_2 \quad \Longrightarrow \quad d_2 = \frac{v^2}{2a}.$$

Part (i)

If $u > 0$ the velocity rises from u to v and stays positive, so distance equals displacement:

$$v^2 = u^2 + 2(3a)d_1 \quad \Longrightarrow \quad d_1 = \frac{v^2 - u^2}{6a}.$$

Hence

$$3d_1 = \frac{v^2 - u^2}{2a} < \frac{v^2}{2a} = d_2,$$

since $u^2 > 0$. So $3d_1 < d_2$, as required.

Part (ii)

Now $u < 0$, and the first stage splits in two.

Ask yourself Why can't we just use $v^2 = u^2 + 6ad_1$ again?

That formula gives the *displacement*, $\frac{v^2-u^2}{6a}$. But the particle moves backwards first. With acceleration $3a$ it slows from speed $|u|$ to rest, covering $\frac{u^2}{6a}$ (from $0 = u^2 - 2(3a)s$). It then speeds up from rest to v, covering $\frac{v^2}{6a}$. The distance is the sum:

$$d_1 = \frac{u^2 + v^2}{6a}, \qquad \text{so} \qquad 3d_1 = \frac{u^2 + v^2}{2a}.$$

Comparing with $d_2 = \frac{v^2}{2a}$ and $2d_2 = \frac{2v^2}{2a}$:

- $3d_1 - d_2 = \dfrac{u^2}{2a} > 0$, since $u \neq 0$;
- $2d_2 - 3d_1 = \dfrac{v^2 - u^2}{2a} > 0$, since $v > |u|$.

Hence $d_2 < 3d_1 < 2d_2$, as required.

Average speed

The times do not care about direction. In stage one the velocity changes by $v - u$ at rate $3a$, and in stage two it changes by v at rate a, so

$$t_1 = \frac{v - u}{3a}, \qquad t_2 = \frac{v}{a}, \qquad t_1 + t_2 = \frac{4v - u}{3a},$$

whichever the sign of u.

Case $u > 0$. The total distance is $\frac{v^2-u^2}{6a} + \frac{3v^2}{6a} = \frac{4v^2-u^2}{6a}$, so the average speed is

$$S_+ = \frac{4v^2-u^2}{6a} \cdot \frac{3a}{4v-u} = \frac{4v^2-u^2}{2(4v-u)}.$$

Case $u < 0$. Write $u = -w$ with $0 < w < v$. The total distance is $\frac{v^2+w^2}{6a} + \frac{3v^2}{6a} = \frac{4v^2+w^2}{6a}$ and the time is $\frac{4v+w}{3a}$, so

$$S_- = \frac{4v^2+w^2}{2(4v+w)}.$$

To compare these, measure each against $\frac{1}{2}v$:

$$S_+ - \tfrac{1}{2}v = \frac{4v^2-u^2-v(4v-u)}{2(4v-u)} = \frac{u(v-u)}{2(4v-u)},$$

$$S_- - \tfrac{1}{2}v = \frac{4v^2+w^2-v(4v+w)}{2(4v+w)} = -\frac{w(v-w)}{2(4v+w)}.$$

Since $0 < u < v$ in the first case, the first expression has a positive numerator and a positive denominator, so $S_+ > \frac{1}{2}v$. Since $0 < w < v$ in the second, the second expression is negative, so $S_- < \frac{1}{2}v$. Therefore

$$S_- < \tfrac{1}{2}v < S_+,$$

and the average speed is greater when $u > 0$, as required. Notice that this holds for any values of $|u|$ in the two cases, not just when they are equal: $\frac{1}{2}v$ separates every $u > 0$ journey from every $u < 0$ journey with the same v and a.

Checks

- Take $v = 2$, $a = 1$. With $u = 1$: $d_1 = \frac{1}{2}$, $d_2 = 2$, so $3d_1 = 1.5 < 2$, and the time is $\frac{7}{3}$, giving $S_+ = \frac{2.5}{7/3} \approx 1.071 > 1$. With $u = -1$: $d_1 = \frac{5}{6}$, so $2 < 3d_1 = 2.5 < 4$, and the time is 3, giving $S_- = \frac{17/6}{3} \approx 0.944 < 1$. A step-by-step numerical simulation of the motion reproduces these values.
- Limits: as $u \to 0$ or $|u| \to v$ both average speeds tend to $\frac{1}{2}v$, which is the average speed of the simple stop-start journey from rest. The inequalities also become equalities at the edges of the allowed range: $3d_1 \to d_2$ as $u \to 0$ in both parts, and $3d_1 \to 2d_2$ as $u \to -v$ in part (ii).

Common slips

- Using $d_1 = \frac{v^2-u^2}{6a}$ in part (ii). That is the displacement, and it would give $3d_1 < d_2$ again, contradicting what you are asked to show.
- Thinking the time also needs splitting when $u < 0$. It doesn't: $v = u + 3at_1$ is valid straight through the turning point.

Beyond STEP

The distinction between displacement and distance is the distinction between $\int v\,\mathrm{d}t$ and $\int |v|\,\mathrm{d}t$. The second is the arc length of the path, and it is what a car's odometer measures. In university mechanics, arc length becomes the natural parameter for describing a curved trajectory, leading to the tangential and normal components of acceleration. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops kinematics in this way.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.